LAB 1: AWS Identity and Access Management (IAM)

AIM

To study and implement AWS IAM by creating users, assigning them to groups, and controlling access using policies.

THEORY

AWS Identity and Access Management (IAM) enables secure control over access to AWS resources. It allows creation of users, groups, and roles, and assignment of permissions through policies. Users represent individuals, groups organize users with similar access, and policies define allowed actions. Using groups simplifies permission management and enforces security by restricting unauthorized access.

PROCEDURE

1. Open AWS Console and navigate to IAM.
2. View existing users and groups.
3. Assign users to groups (S3-Support, EC2-Support, EC2-Admin).
4. Login using each user account.
5. Verify access permissions for S3 and EC2 services.
6. Submit lab and check results.

RESULT

Successfully implemented IAM by assigning users to appropriate groups and verifying access control. It was observed that permissions were enforced correctly based on assigned policies.

LAB 2: Build VPC and Launch a Web Server

AIM

To create a Virtual Private Cloud (VPC), configure subnets and routing, set up security groups, and launch an EC2 instance as a web server.

THEORY

Amazon VPC allows creation of a logically isolated network in AWS. It consists of subnets, route tables, internet gateway, and NAT gateway. Public subnets provide direct internet access, while private subnets are secured and use NAT for outbound connectivity. Security groups act as firewalls controlling traffic, and EC2 instances serve as virtual servers.

PROCEDURE

1. Create VPC with CIDR 10.0.0.0/16.
2. Create public and private subnets.
3. Add additional subnets in another availability zone.
4. Configure route tables (IGW for public, NAT for private).
5. Create security group allowing HTTP access.
6. Launch EC2 instance in public subnet.
7. Add user data script to install web server.
8. Access website using public DNS.

RESULT

Successfully created a VPC with public and private subnets, configured routing and security, and deployed an EC2 instance running a web server accessible via the internet.